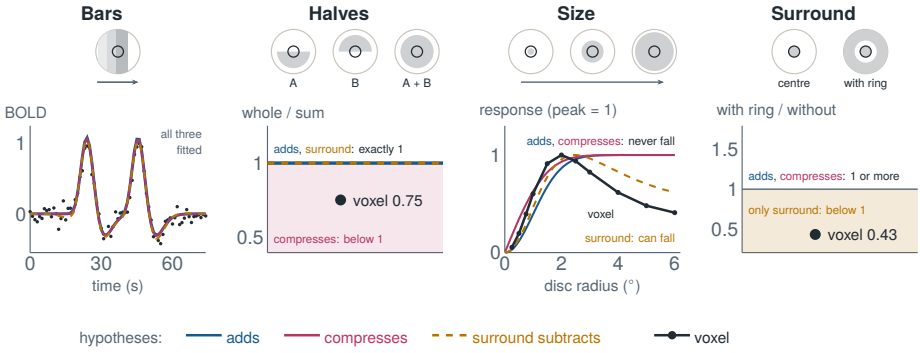
POPYLAR PRF WORKSHOP · COIMBRA · FRIDAY 6 NOVEMBER 2026

# ADVANCED PRF MODELS

## Where, what, how and when: from population receptive fields to richer encoding models

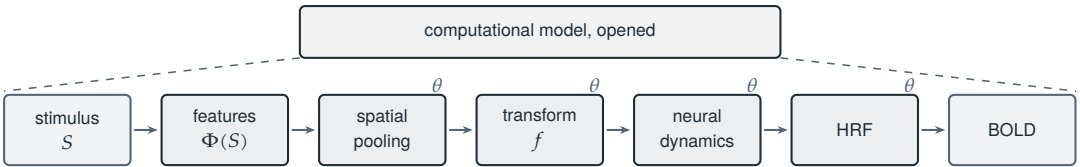
Tomas Knapen

# EXPLAINING YESTERDAY'S VOXEL



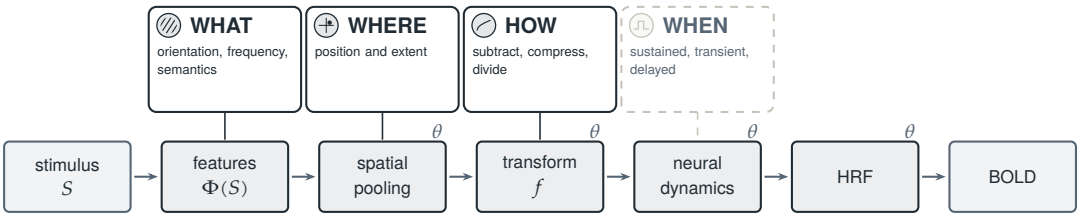

- We designed the experiment; now we have to explain the response.

# EVERY PRF IS AN ENCODING MODEL




- A pRF model is a map from stimulus to predicted measurement, with parameters to fit and to test.

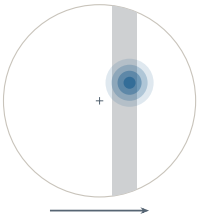
# WHERE, WHAT, HOW, WHEN

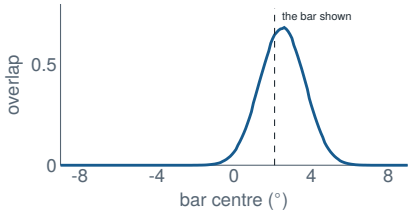



Four questions about one model:

- where it pools
- what it is tuned to
- how it transforms
- when it integrates

# A GAUSSIAN SUMS OVER SPACE

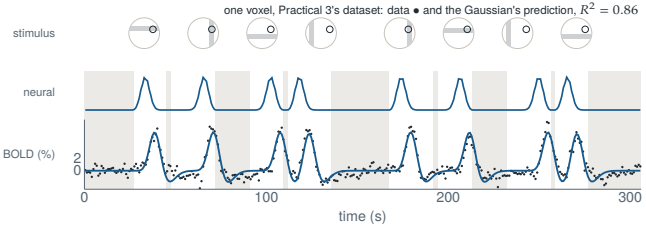





$$r(t) = \sum_{x,y} G(x, y;\, x_0, y_0, \sigma)\, S(x, y, t)$$

- The response is a weighted sum of stimulation over space; the weights are a Gaussian. The bar's aperture is the stimulus $S$: one where it is, zero elsewhere.

# FROM STIMULUS TO BOLD




$$r(t) = \sum_{x,y} G(x, y;\, x_0, y_0, \sigma)\, S(x, y, t), \qquad \mathrm{BOLD}(t) = (h * r)(t)$$

- One set of parameters predicts the whole time course; that prediction is what we fit.

# WHAT THE GAUSSIAN FORBIDS

**Parts add**




$r(A+B) = r(A) + r(B)$

**More is never less**



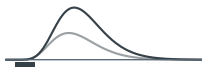

every weight positive:
added stimulation adds

**Only where counts**




same aperture,
same response

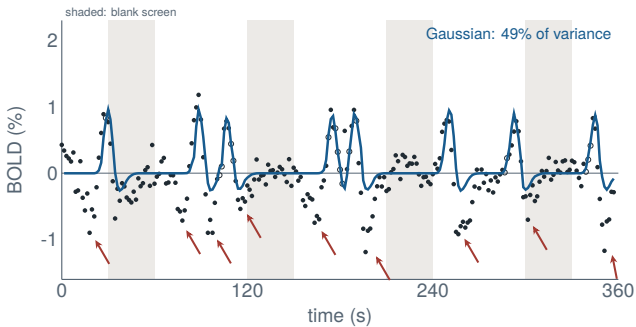
**Activations sum in time**

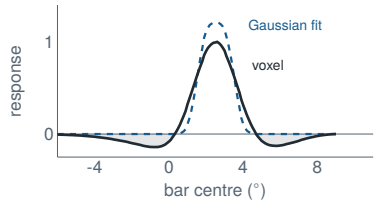


twice as long,
twice the response

- All models are wrong, but some are useful *(Box 1979)*: useful because they forbid things; each forbidden response is a test.

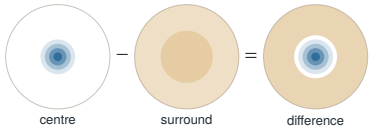
# MORE STIMULUS, LESS RESPONSE




- A bar beside the centre can push the response below baseline; no positive-weighted sum does that *(Zuiderbaan et al. 2012)*.

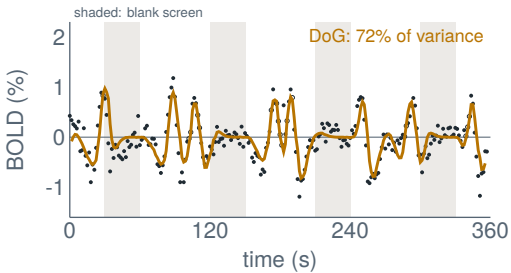
# CENTRE MINUS SURROUND

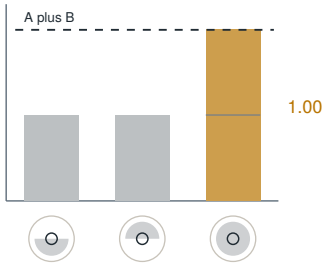



$$r(t) = \sum_{x,y} \left[ a_c \, G_c(x,y) - a_s \, G_s(x,y) \right] S(x,y,t)$$

- DoG subtracts a broader Gaussian: stimulation around the centre now lowers the response.

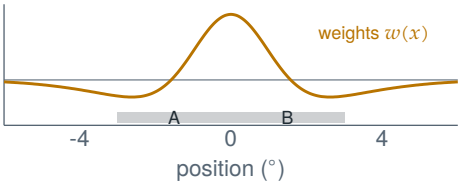
# A SURROUND, BUT STILL A SUM




- DoG captures the negative lobe but still adds: the whole equals the sum of its parts.

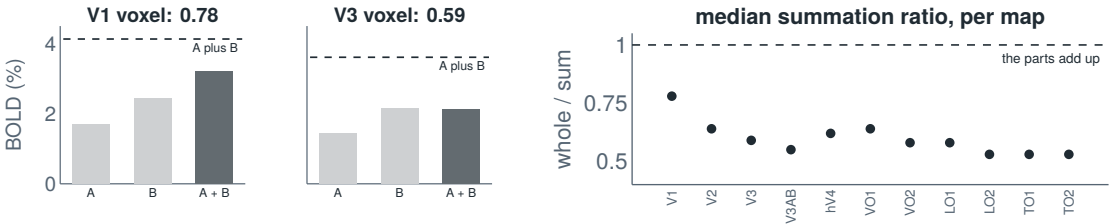
# ANY LINEAR FILTER ADDS




$$r(A) = \sum_x w(x)\, A(x)$$
$$r(B) = \sum_x w(x)\, B(x)$$
$$r(A{+}B) = \sum_x w(x)\, [A(x) + B(x)]$$
$$= r(A) + r(B)$$

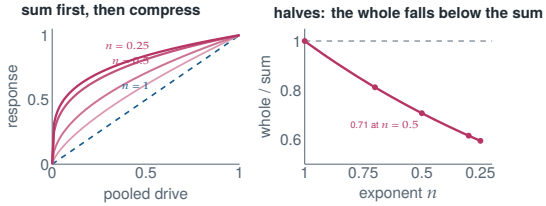
- For any weights $w$, positive or negative:
  $r(A + B) = \sum w\,(A + B) = \sum wA + \sum wB = r(A) + r(B).$

# THE WHOLE IS LESS THAN THE SUM




- Yesterday's experiment is today's constraint: no linear model falls below the sum *(Kay et al. 2013)*.

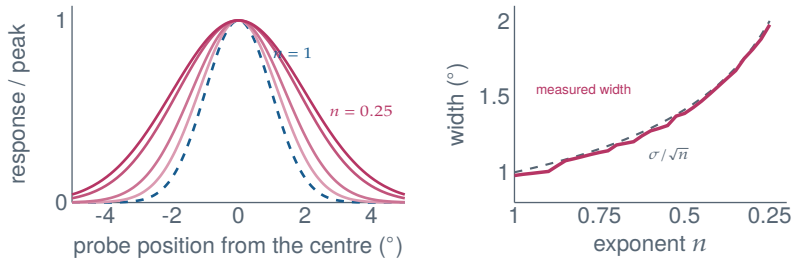
# SUM FIRST, THEN COMPRESS




$$r(t) = \Big[ \sum_{x,y} G(x,y)\, S(x,y,t) \Big]^{n}, \qquad 0 < n \leq 1$$

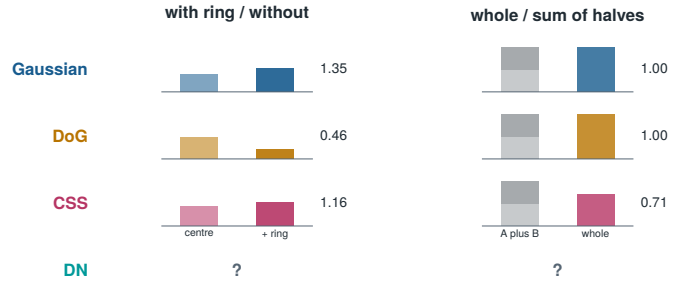
- CSS raises the summed drive to a power below one; the whole then falls below the sum.

# COMPRESSION WIDENS THE FIELD



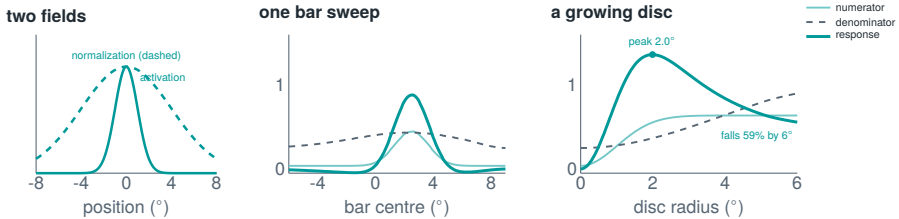

- With $n < 1$ the response to a small probe falls off more slowly than the Gaussian, as $G^n$: the effective size is $\sigma / \sqrt{n}$, larger than $\sigma$ *(Kay et al. 2013)*.

# TWO FAILURES, TWO FIXES




- DoG gives the negative lobe, CSS the compression; neither gives both.

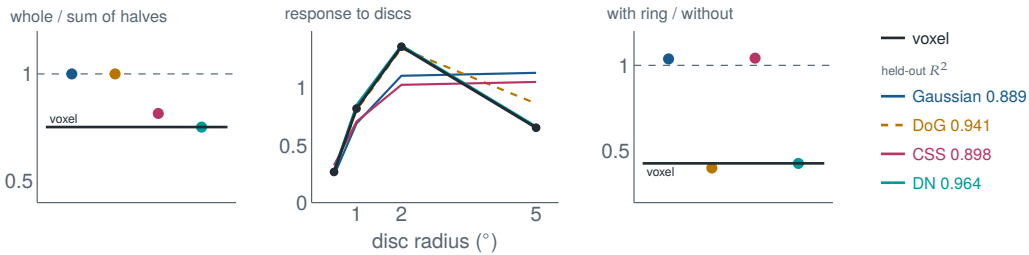
# DIVIDE BY THE CONTEXT




$$r = \frac{a\,A(S) + b}{c\,N(S) + d} - \frac{b}{d}$$

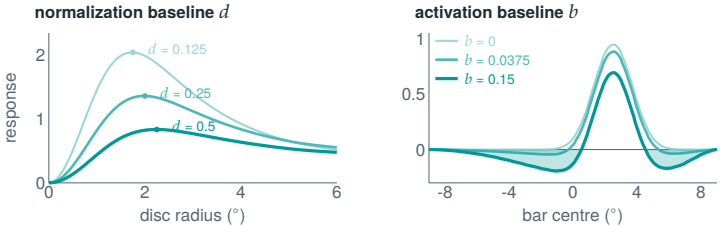
- A canonical computation *(Carandini and Heeger 2012)*: a small field's drive over a larger field's.

# ONE OPERATION, BOTH SIGNATURES




- One divisive operation fits suppression and compression in the same voxel, as in measured voxels *(Aqil et al. 2021)*.

# PARAMETERS SHAPE RESPONSES




$$r = \frac{a\,A(S) + b}{c\,N(S) + d} - \frac{b}{d}$$

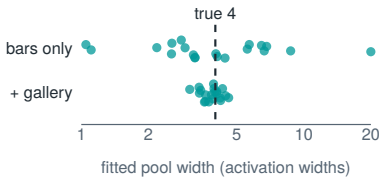
- $d$ sets the oversaturation and the compression; $b$ sets the surround.

# PINNING THE PARAMETERS

Scaling $a$, $b$, $c$ and $d$ by any factor $k$ leaves the prediction unchanged.

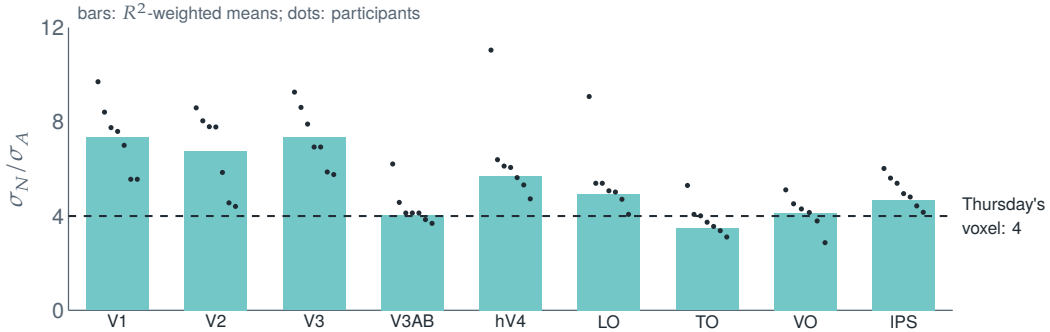
$$\frac{ka\,A + kb}{kc\,N + kd} - \frac{kb}{kd} = \frac{aA + b}{cN + d} - \frac{b}{d}$$

any factor $k$: same prediction



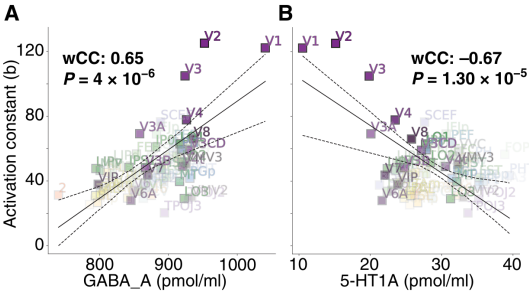

- Fix one scale ($c = 1$); probe the pool with stimuli that reach it.

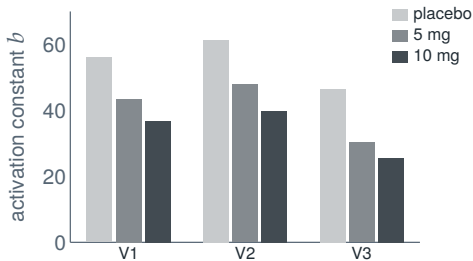
# ACROSS THE VISUAL HIERARCHY




- The normalization pool's width changes systematically across visual areas *(Aqil et al. 2021)*.

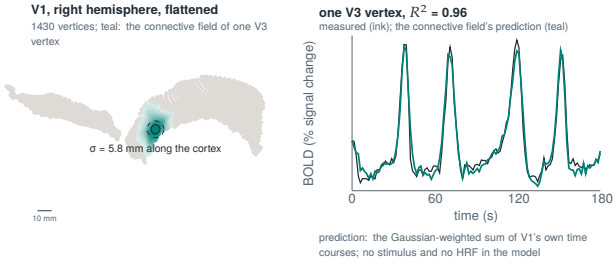
# FROM PARAMETERS TO BIOLOGY



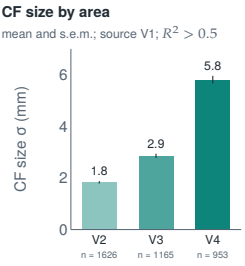

- $b$ tracks $GABA_A$ and $5\text{-}HT_{1A}$ receptor densities across areas *(Aqil et al. 2024)*; psilocybin lowers it *(Aqil et al. 2025)*.

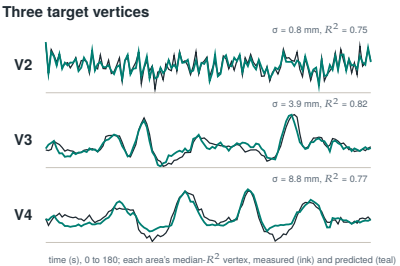
# CONNECTIVE FIELDS: PRFS ON THE CORTEX




- A Gaussian over cortical distance, fed by V1's activity *(Haak et al. 2013)*.

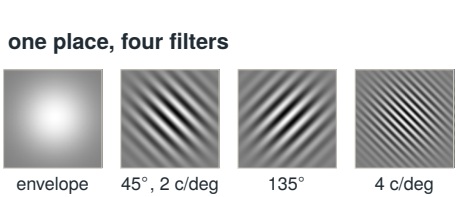
# FIELDS GROW UP THE HIERARCHY

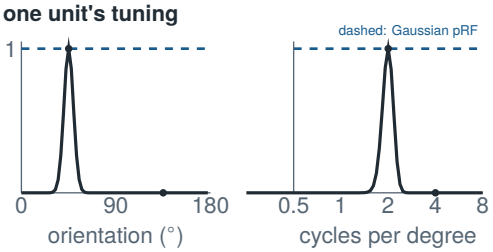



- 1.8 mm in V2, 2.9 in V3, 5.8 in V4: each area pools a wider patch of V1.

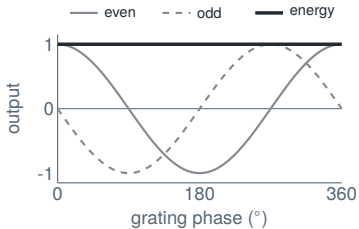
# SAME PLACE, OTHER FEATURES

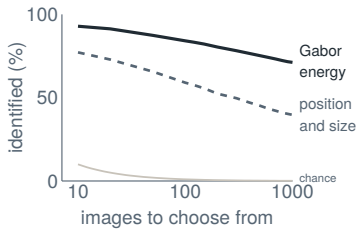



- Two images at one place can drive a voxel differently: position is not the whole stimulus.

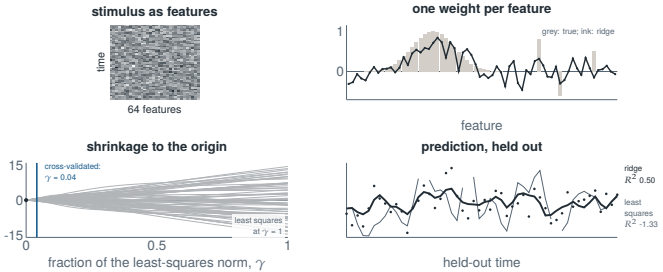
# A RECEPTIVE FIELD FOR IMAGES



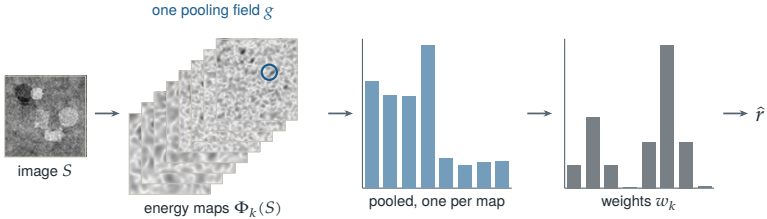

- Two Gabor filters in quadrature, squared and summed, respond to contrast whatever its phase. Gabor energy pooled over space predicted responses to new photographs, well enough to identify them: 92% and 72% of 120 new images in two subjects, where chance is 0.8% *(Kay et al. 2008)*.

# THOUSANDS OF FEATURES




- The weights become the description *(Huth et al. 2016)*; ridge shrinks them toward the origin *(Rokem and Kay 2020)*.
- A different approach from the few-parameter process models in prfmodel.

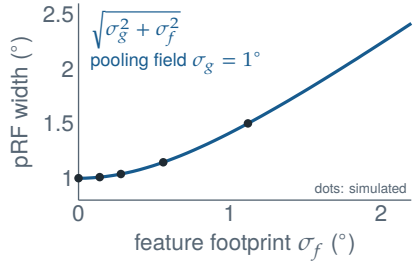
# THE FEATURE-WEIGHTED RF




$$\hat{r} = \sum_k w_k \sum_{x,y} g(x, y;\, \theta)\, \Phi_k(S, x, y)$$

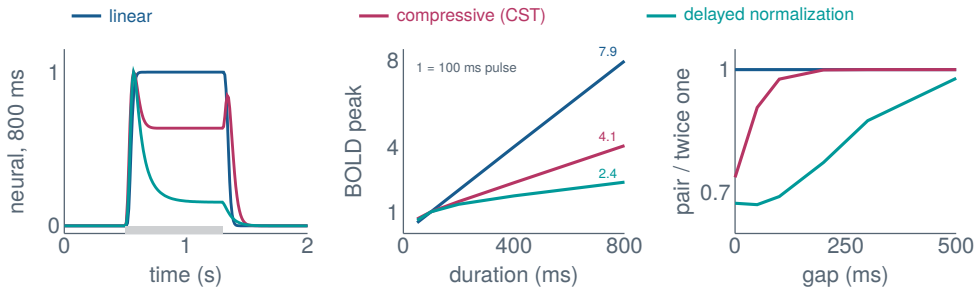
- One shared pooling field, features as rich as needed: where times what *(St-Yves and Naselaris 2018)*.

# POOLING WIDTH IN VISUAL SPACE



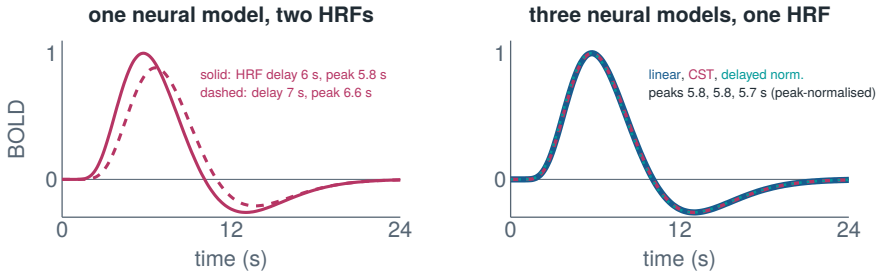

- The field pools feature maps whose filters have their own footprint $\sigma_f$: in visual degrees the pRF is $\sqrt{\sigma_g^2 + \sigma_f^2}$, so the pooling field alone understates its size *(St-Yves and Naselaris 2018)*.

# SAME PLACE, OTHER TIMING



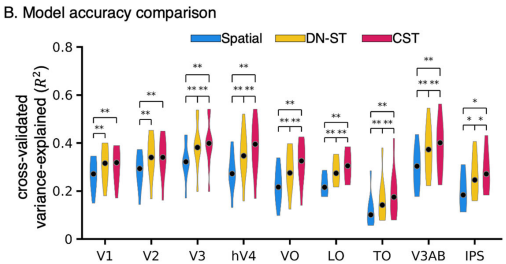

- Same place, same features, different timing: space alone does not set the response.

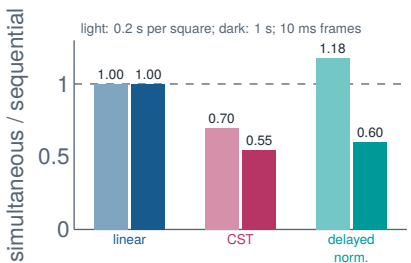
# THE HRF IS NOT NEURAL TIME




- A slower HRF moves the BOLD peak for every stimulus alike (5.8 to 6.6 s here); the neural models, through one HRF, give the same BOLD shape and differ in how the amplitude grows with duration, which a linear HRF cannot mimic.

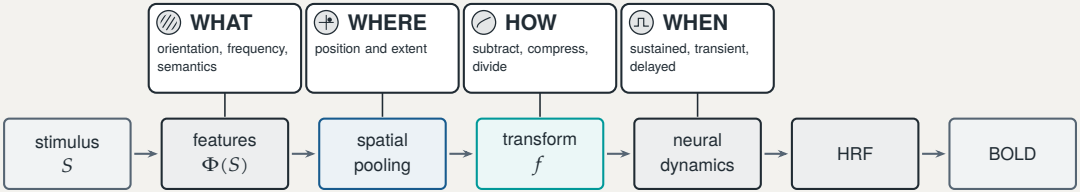
# SPATIOTEMPORAL PRFS




- Spatiotemporal pRF models exist and fit fMRI *(Kim et al. 2024; Kupers et al. 2024)*; prfmodel ships two in its beta release.

# MODELS FOR THE QUESTION




1. Specify a predictive hypothesis.
2. Design stimuli that constrain its parameters.
3. Compare predictions on data not used to fit it.

Practical 3, today at 12:00

Gaussian — DoG — CSS — DN

- prfmodel implements these models and their components; time to play with them in practice.

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Aqil, Marco, Tomas Knapen, and Serge O. Dumoulin. 2021. “Divisive Normalization Unifies Disparate Response Signatures Throughout the Human Visual Hierarchy.” *Proceedings of the National Academy of Sciences* 118 (46): e2108713118. https://doi.org/10.1073/pnas.2108713118.

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